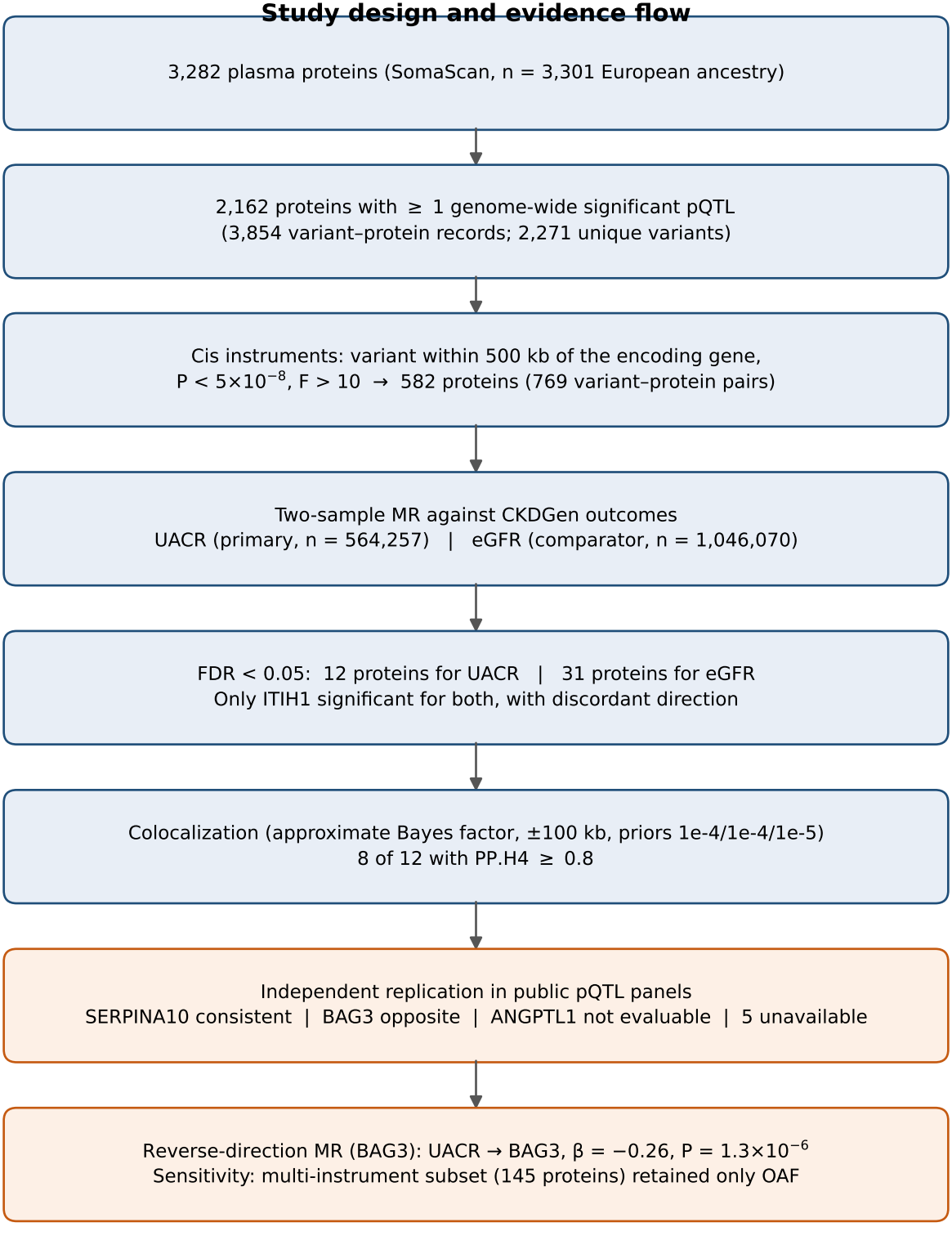
Study design and evidence flow
3,282 plasma proteins (SomaScan, n = 3,301 European ancestry)
2,162 proteins with ≥ 1 genome-wide significant pQTL
(3,854 variant–protein records; 2,271 unique variants)
Cis instruments: variant within 500 kb of the encoding gene,
P < 5×10−8, F > 10 → 582 proteins (769 variant–protein pairs)
Two-sample MR against CKDGen outcomes
UACR (primary, n = 564,257) | eGFR (comparator, n = 1,046,070)
FDR < 0.05: 12 proteins for UACR | 31 proteins for eGFR
Only ITIH1 significant for both, with discordant direction
Colocalization (approximate Bayes factor, ±100 kb, priors 1e-4/1e-4/1e-5)
8 of 12 with PP.H4 ≥ 0.8
Independent replication in public pQTL panels
SERPINA10 consistent | BAG3 opposite | ANGPTL1 not evaluable | 5 unavailable
Reverse-direction MR (BAG3): UACR → BAG3, β = −0.26, P = 1.3×10−6
Sensitivity: multi-instrument subset (145 proteins) retained only OAF